

Thermodynamic State Vector Interface and Exergy Auditing in the Web of Life Simulation Engine (Sprint 010)

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Abstract

Ecosystem simulation models frequently struggle to reconcile macroscopic biogeochemical cycles with rigorous thermodynamic constraints. In Sprint 010, we establish strict mathematical and programmatic contracts for thermodynamic state vectors within the **Web of Life** simulation engine (<https://github.com/pascalranoroarijaona/WebOfLife>). Treating the Earth Pod as an open thermodynamic system, we formalize the First and Second Laws of Thermodynamics, tracking internal energy conservation, irreversible entropy generation ($\dot{S}_{\text{gen}}$), and exergy destruction ($\dot{I} = T_0 \dot{S}_{\text{gen}}$). Through strict TypeScript interface contracts and runtime validation guards, our architecture guarantees adherence to the Clausius inequality, bridging systems ecology with non-equilibrium thermodynamics.

1 Introduction and Systems Ecology Framing

The **Web of Life** engine models Earth’s biosphere as a coupled dynamical system driven by solar exergy inflow and sustained by thermodynamic dissipation. Living systems operate as far-from-equilibrium dissipative structures that maintain internal order by exporting entropy to their external environment.

Sprint 010 formalizes these principles into executable software contracts located in `src/thermodynamics/ty` and `src/thermodynamics/thermodynamic_structure.ts`.

2 Thermodynamic Foundations

2.1 First Law of Thermodynamics (Energy Conservation)

For the Earth Pod open thermodynamic system, energy conservation over time step dt is governed by:

$$\frac{dU_{\text{sys}}}{dt} = \Phi_{\text{solar}} - \Phi_{\text{thermal}} - \Phi_{\text{sensible}} - \Phi_{\text{latent}} + \sum_k \dot{W}_k \quad (1)$$

where mass is conserved ($\sum \dot{m} = 0$) and external energy input is restricted to solar radiation flux (Φ_{solar}).

2.2 Second Law of Thermodynamics & Exergy Destruction

The entropy change of the system is governed by:

$$\frac{dS_{\text{sys}}}{dt} = \sum_k \frac{\dot{Q}_k}{T_k} + \dot{S}_{\text{gen}} \quad (2)$$

where $\dot{S}_{\text{gen}} \geq 0$ represents the internal irreversible entropy generation rate.

The **Exergy Destruction Rate** ($\dot{I}$) quantifies thermodynamic irreversibility relative to the ambient reference temperature T_0 :

$$\dot{I} = T_0 \dot{S}_{\text{gen}} \geq 0 \quad [\text{W}] \quad (3)$$

3 Interface Contracts & Implementation

The TypeScript interface contracts established in `src/thermodynamics/types.ts` enforce strict type safety and runtime verification:

```
export interface BoundaryFluxVector {
  solarRadiationIn: number; // [W] Incoming solar shortwave flux
  thermalRadiationOut: number; // [W] Outgoing longwave thermal radiation
  sensibleHeatFlux: number; // [W] Convective/conductive boundary sensible heat
  latentHeatFlux: number; // [W] Evapotranspirative latent heat flux
  netMassEnthalpyFlux: number; // [W] Net enthalpy carried by boundary mass transfers
}

export interface ThermodynamicStateVector {
  timestamp: number; // [s] Simulation epoch time
  ambientTemperature: number; // T_0 [K] Reference ambient temperature
  systemTemperature: number; // T [K] Effective internal system temperature
  internalEnergy: number; // U [J] Total internal energy of system stocks
  totalEntropy: number; // S [J/K] Total system entropy
  entropyGenerationRate: number; // S_gen_dot [W/K] Internal entropy generation rate
  exergyDestructionRate: number; // I_dot [W] Exergy destruction rate (T_0 * S_gen_dot)
  boundaryFluxes: BoundaryFluxVector;
}
```

4 Verification and Validation Guards

Automated test suites assert that $\dot{S}_{\text{gen}} \geq 0$ across diverse metabolic states. Any simulated condition resulting in negative entropy generation triggers a critical system halt:

```
if (state.entropyGenerationRate < -EPSILON) {
  throw new Error('CRITICAL THERMODYNAMIC VIOLATION: S_gen_dot < 0');
}
```

5 Conclusion

Sprint 010 successfully embeds non-equilibrium thermodynamics into the core architecture of the Web of Life engine. Future developments will utilize these exergy metrics to evaluate ecosystem health, trophic efficiency, and evolutionary adaptation.

References

Web of Life Repository: <https://github.com/pascalranoroarijaona/WebOfLife>